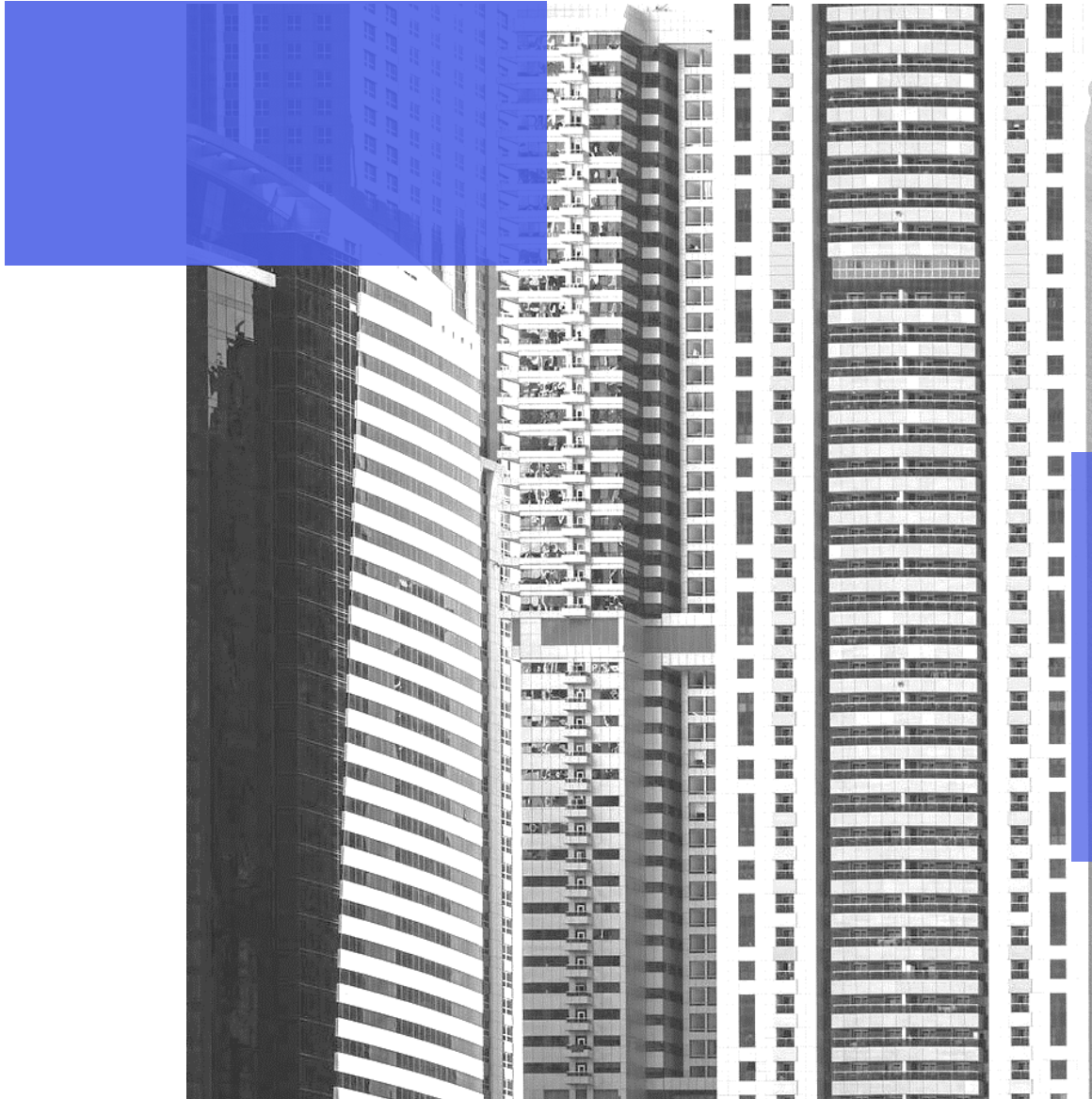


ABDULLAH IHSAN TASLIM

PREDICTING LAPTOP PRICES

“WHAT REALLY DRIVES THE MARKET?”





AGENDA

INTRODUCTION

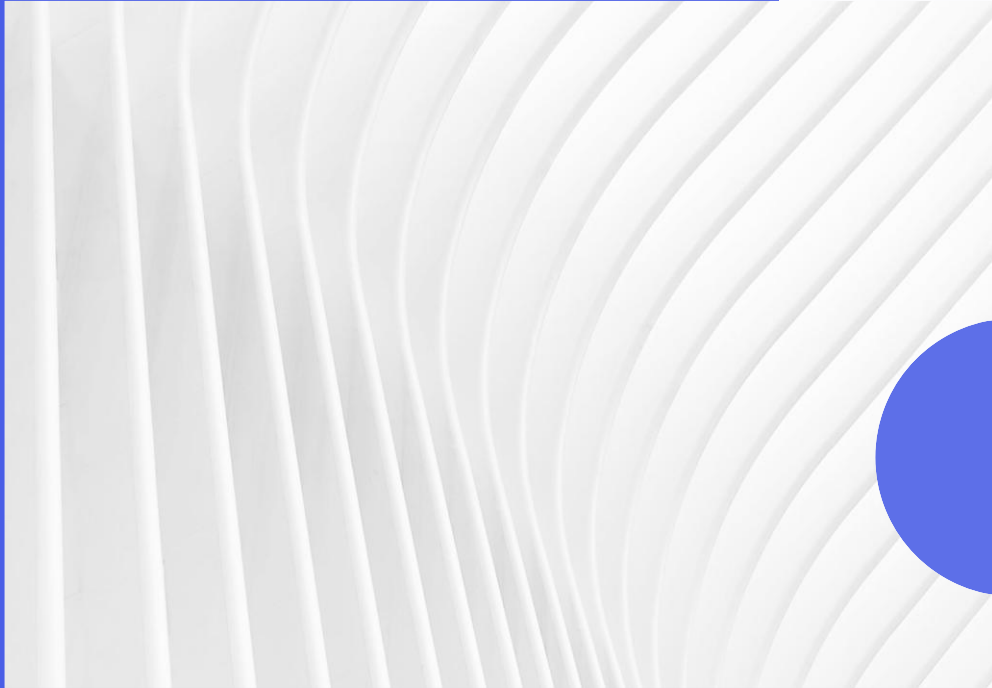
THE MARKET LANDSCAPE

DOMINANT FEATURES

MODEL SPECIFICATION

BUSINESS VALUE

INTRO

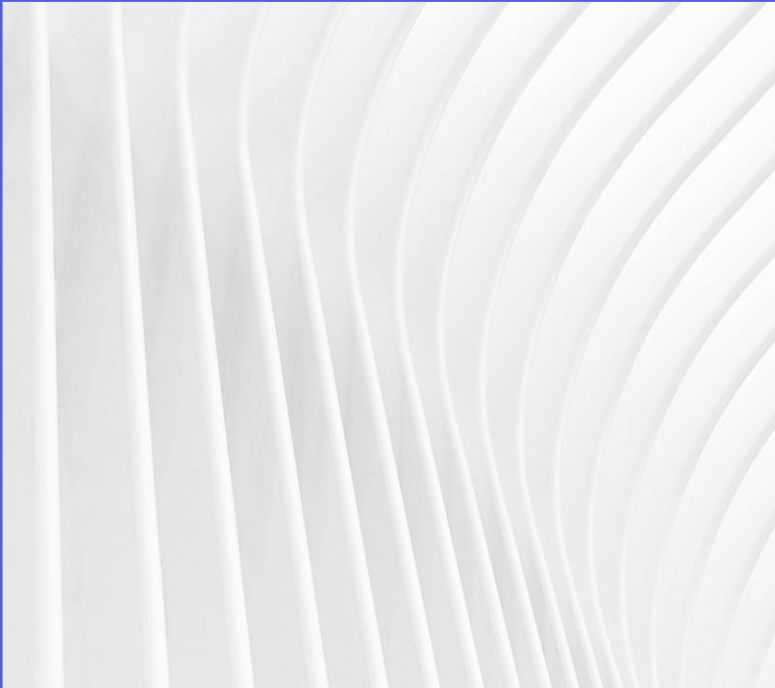


The aim of this study is to analyze dataset that contains detailed information about **1,302 laptops** from various manufacturers. And Implementing ETL, EDA and Modelling in order to answer these question:

*"What features contributes to **price** increase?"*

*"Can a model be made to predict laptop price **accurately**?"*

DATA DESCRIPTION



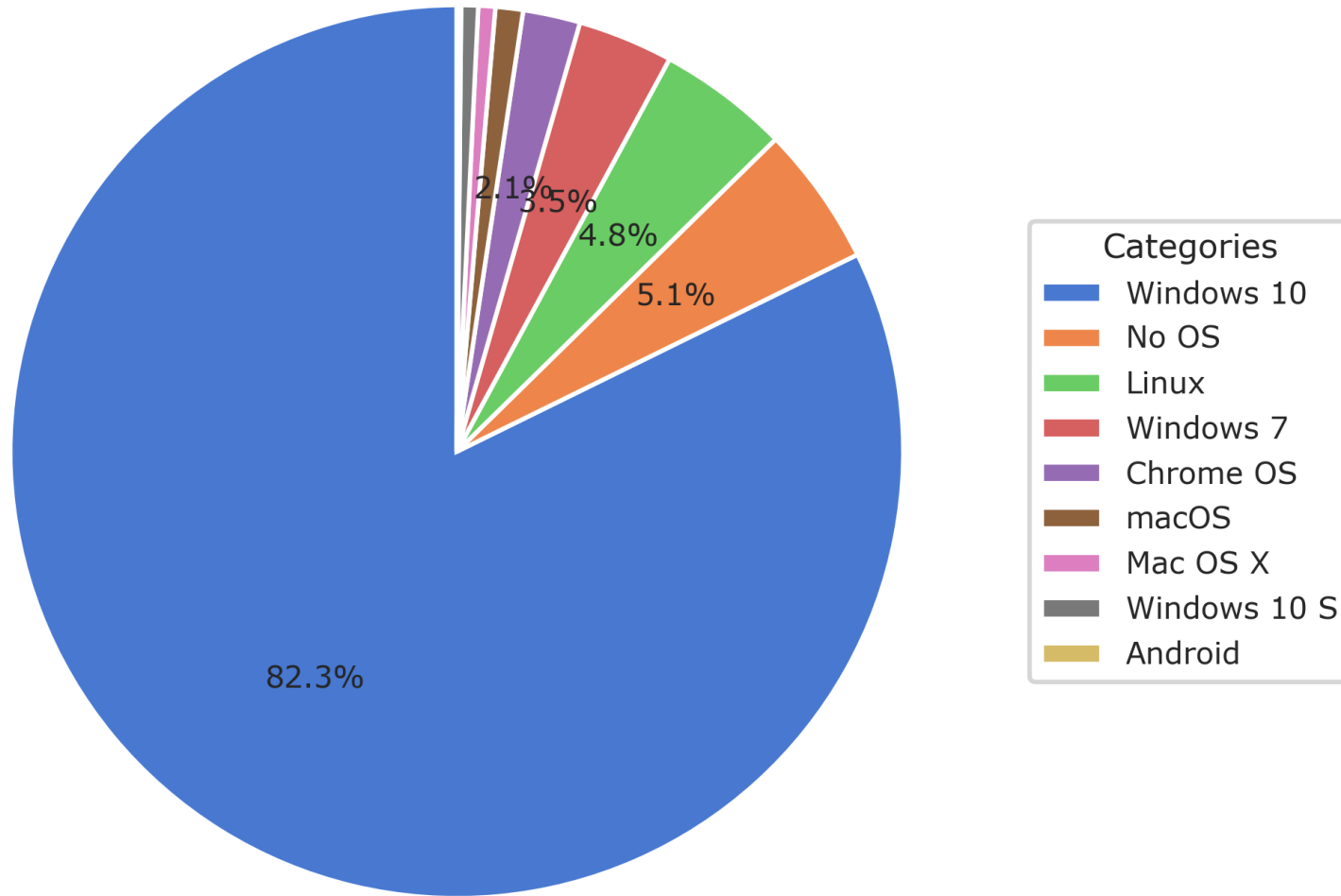
Feature	Description
Unnamed: 0	Unique identifier/index
Company	Laptop manufacturer (Apple, HP, Dell, Lenovo, etc.)
TypeName	Laptop category (Notebook, Ultrabook, Gaming, Workstation, etc.)
Ram	Installed RAM size (GB)
Weight	Laptop weight
Price	Laptop price (Target Variable)
TouchScreen	Indicates whether the laptop has a touchscreen (0/1)
Ips	Indicates IPS display technology (0/1)
ppi	Pixels per inch (screen pixel density)
Cpu Brand	Processor manufacturer/brand
HDD	Hard Disk Drive storage capacity (GB)
SSD	Solid State Drive storage capacity (GB)
Gpu Brand	Graphics card manufacturer
os	Operating System category



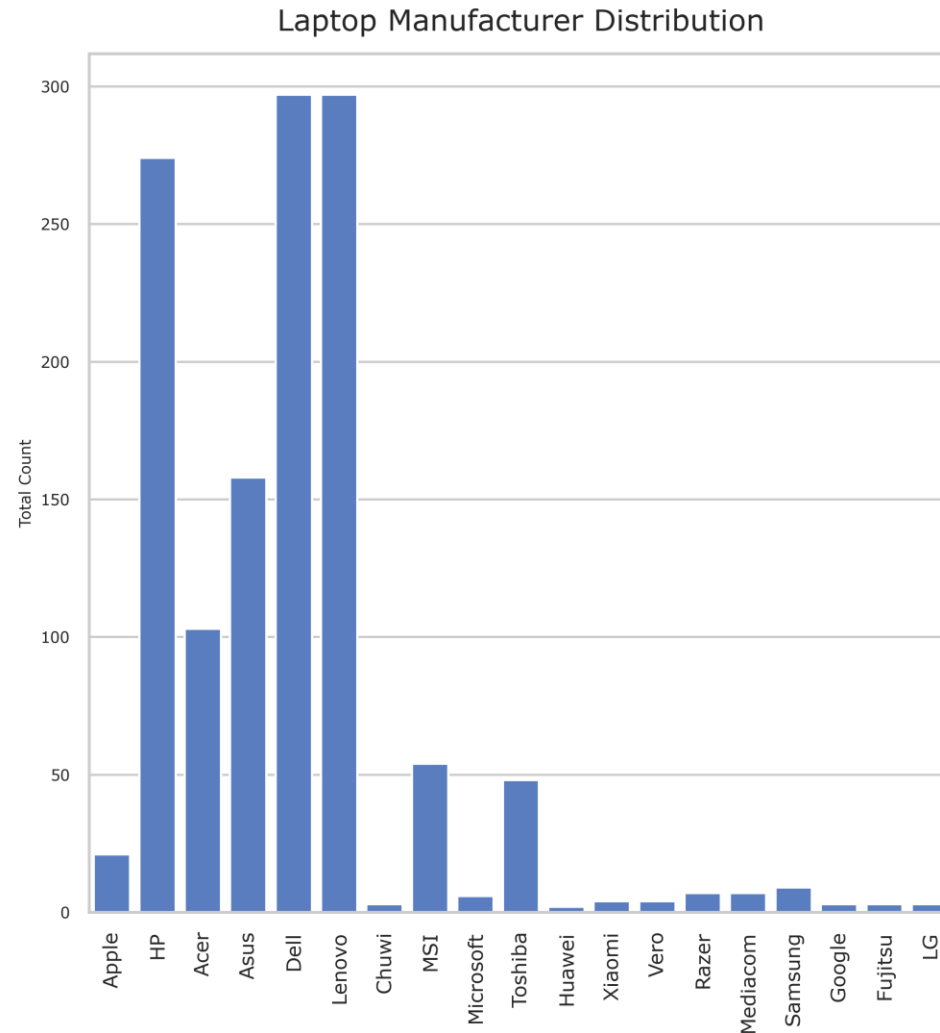
THE MARKET LANDSCAPE

DATA FINDINGS BASED ON THE DATASET

MICROSOFT DOMINATES THE ECOSYSTEM WITH **WINDOWS 10**



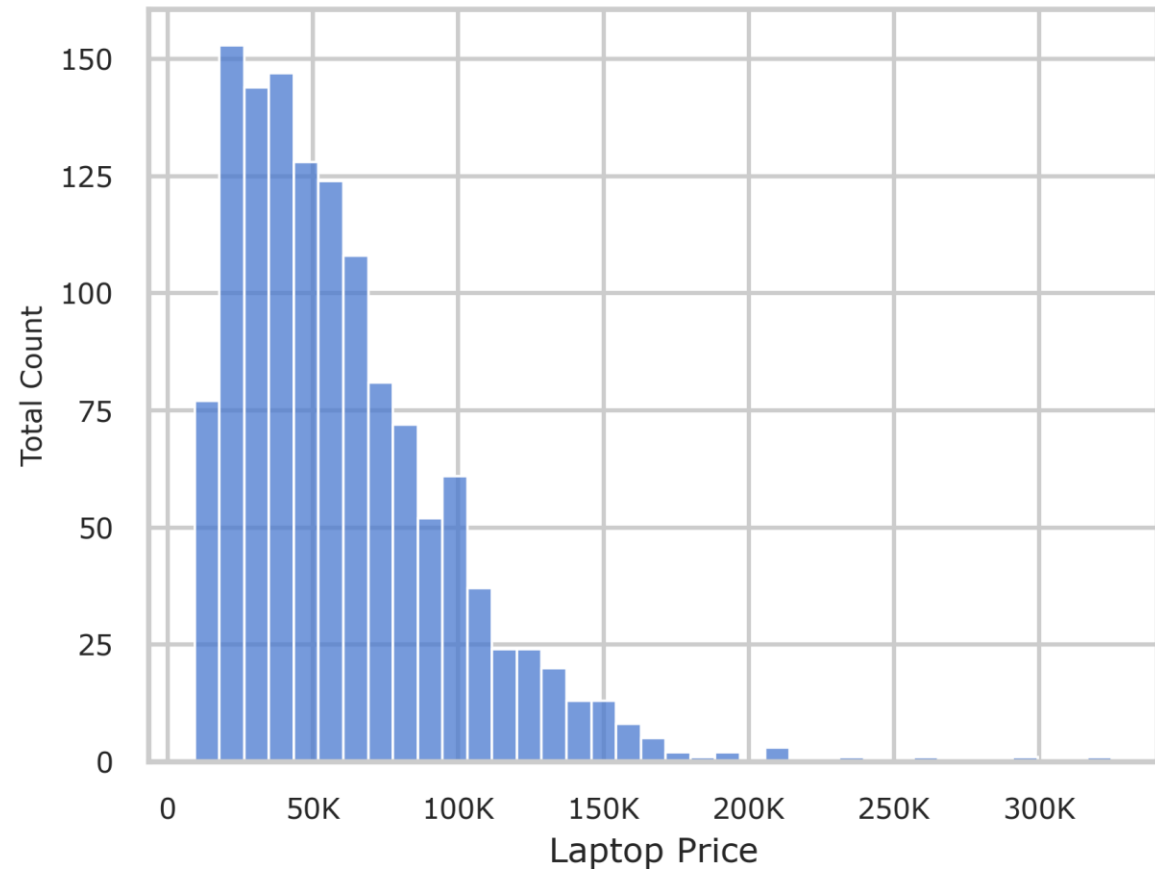
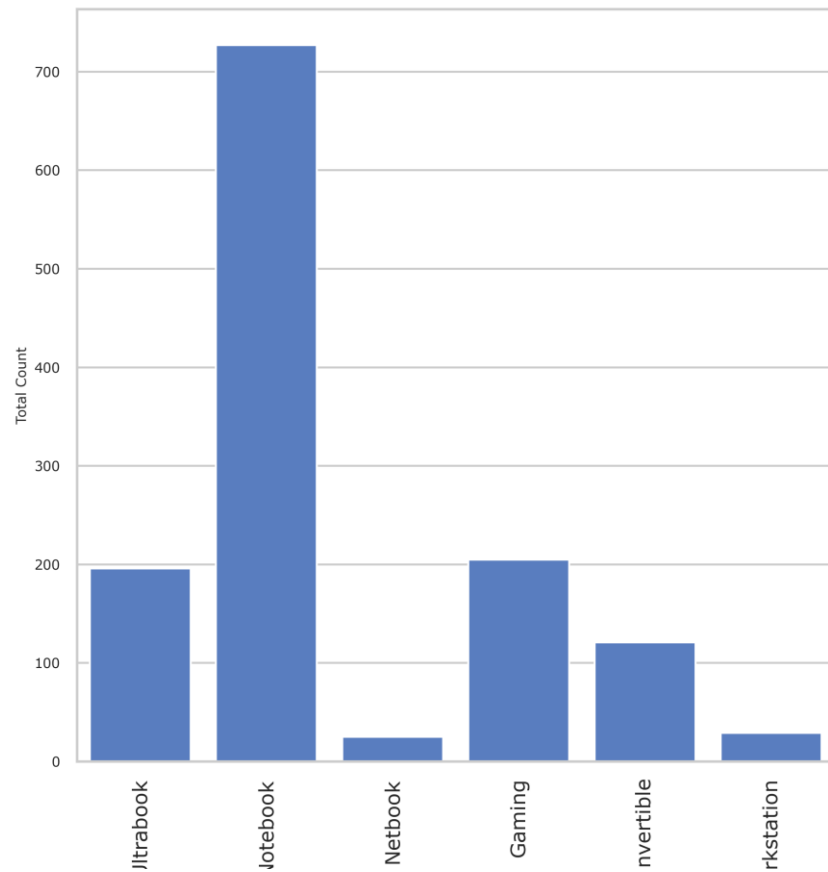
MEAWHILE **LENOVO, DELL & HP** DOMINATE THE HARDWARE



LOWER END DEVICE, NOTEBOOK PRICES AROUND 50K SATURATING THE MARKET

Laptop Price Distribution

Laptop Type Distribution

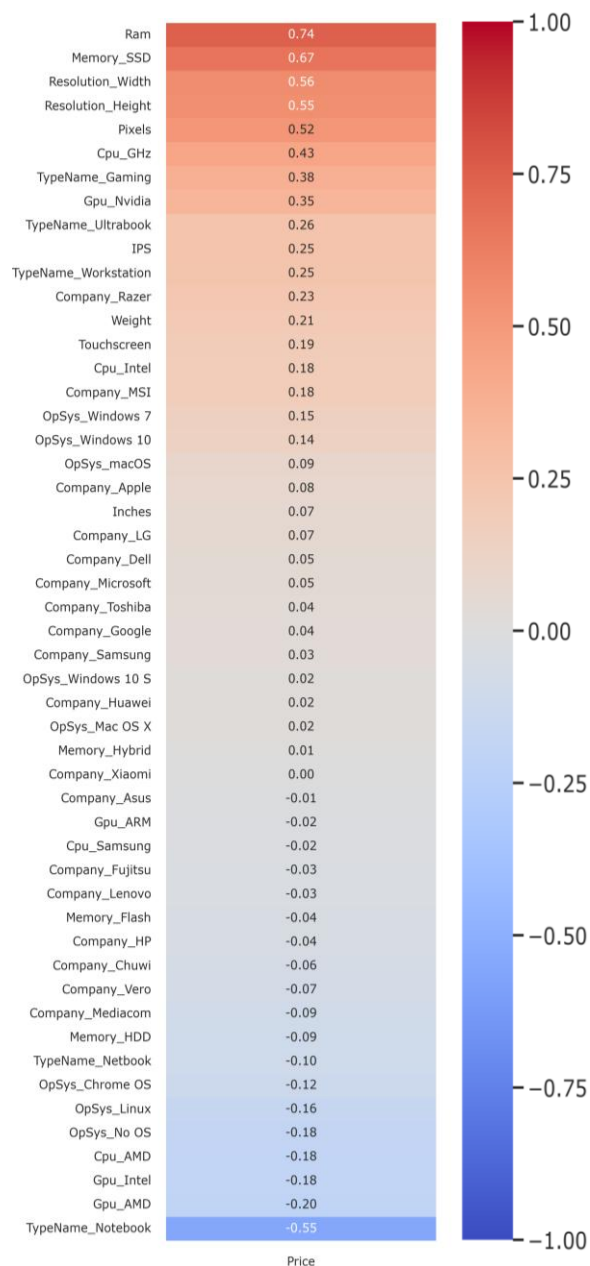


“

THUS, WHAT IS THE **FEATURES** THAT DICTATE
THE PRICES?

”





RAM & SSD ARE THE DICTATORS OF THE PRICE

Feature	Correlation with Price
RAM	0.74
SSD	0.67
Screen size (Pixels)	0.52
CPU GHz	0.43
Nvidia GPU	0.35
Laptop Type (Gaming & Notebook)	(Gaming=0.38 & Notebook=-0.55)

- * The closer it gets to 1 & -1 the stronger it is at affecting price
- * There is also "gaming premium" that indicate specialized, pricier hardware (as noted by Nvidia GPU & Gaming Laptop Type)

THESE FEATURES ARE WHAT'S GOING TO SHAPE THE MODEL

Feature	Correlation with Price
RAM	0.74
SSD	0.67
Screen size (Pixels)	0.52
CPU GHz	0.43
Nvidia GPU	0.35
Laptop Type (Gaming & Notebook)	(Gaming=0.38 & Notebook=-0.55)

* Using a threshold of correlation point > 0.3 & < -0.3 to get moderate to strong correlation with price

THE PREDICTION MODEL USING MULTIPLE LINEAR REGRESSION (SCALED)

$$\text{Price} = (59870.0429) + (14571.1575 * \text{Ram}) + (8593.9328 * \text{Memory_SSD}) + (5650.7677 * \text{Pixels}) + (5124.6021 * \text{Cpu_GHz}) + (123.1021 * \text{Gpu_Nvidia}) + (1909.7656 * \text{TypeName_Gaming}) + (-585.7866 * \text{TypeName_Netbook}) + (-5049.0506 * \text{TypeName_Notebook}) + (2943.9682 * \text{TypeName_Ultrabook}) + (6110.8416 * \text{TypeName_Workstation})$$

	Coefficient	Std. Error	t-value	P > t
const	59870.04	527.34	113.53	0.0
Ram	14571.16	795.9	18.31	0.0
Memory_SSD	8593.93	757.02	11.35	0.0
Pixels	5650.77	622.17	9.08	0.0
Cpu_GHz	5124.6	607.19	8.44	0.0
Gpu_Nvidia	123.1	728.07	0.17	0.8658
TypeName_Gaming	1909.77	1001.05	1.91	0.0566
TypeName_Netbook	-585.79	584.73	-1.0	0.3166
TypeName_Notebook	-5049.05	979.73	-5.15	0.0
TypeName_Ultrabook	2943.97	789.93	3.73	0.0002
TypeName_Workstation	6110.84	614.16	9.95	0.0

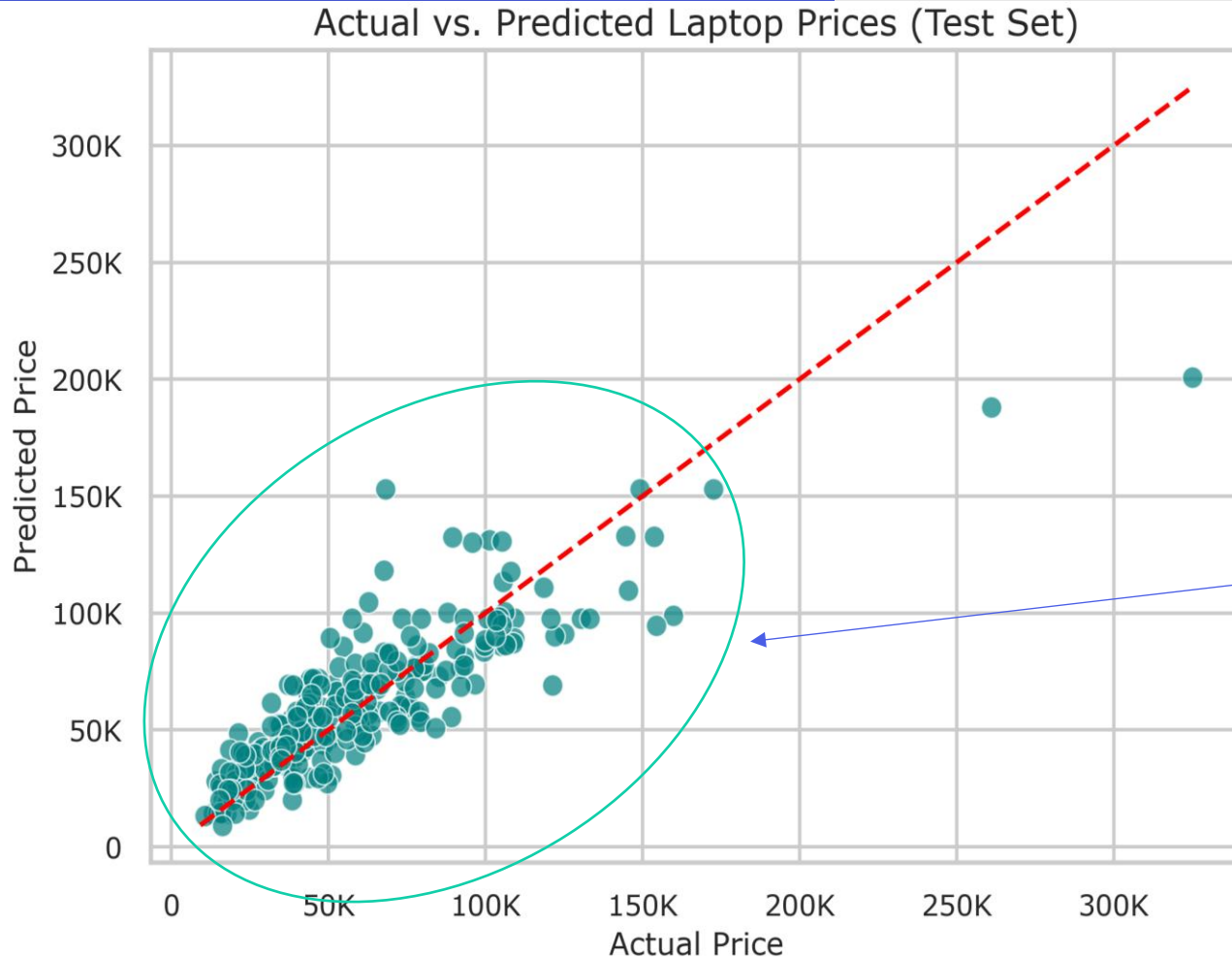
UPGRADING RAM ALMOST COST DOUBLE THAN BETTER SCREEN OR CPU

Variable summary of the Prediction Model

	Coefficient	Std. Error	t-value	P > t
const	60101.9	589.13	102.02	0.0
Ram	14395.82	890.0	16.18	0.0
Memory_SSD	8000.63	843.8	9.48	0.0
Pixels	5349.26	692.96	7.72	0.0
Cpu_GHz	5362.17	678.56	7.9	0.0
Gpu_Nvidia	681.79	840.41	0.81	0.4174
TypeName_Gaming	472.11	1148.3	0.41	0.6811
TypeName_Netbook	-949.52	611.27	-1.55	0.1206
TypeName_Notebook	-6689.21	1130.41	-5.92	0.0
TypeName_Ultrabook	2202.39	891.76	2.47	0.0137
TypeName_Workstation	5212.63	684.73	7.61	0.0

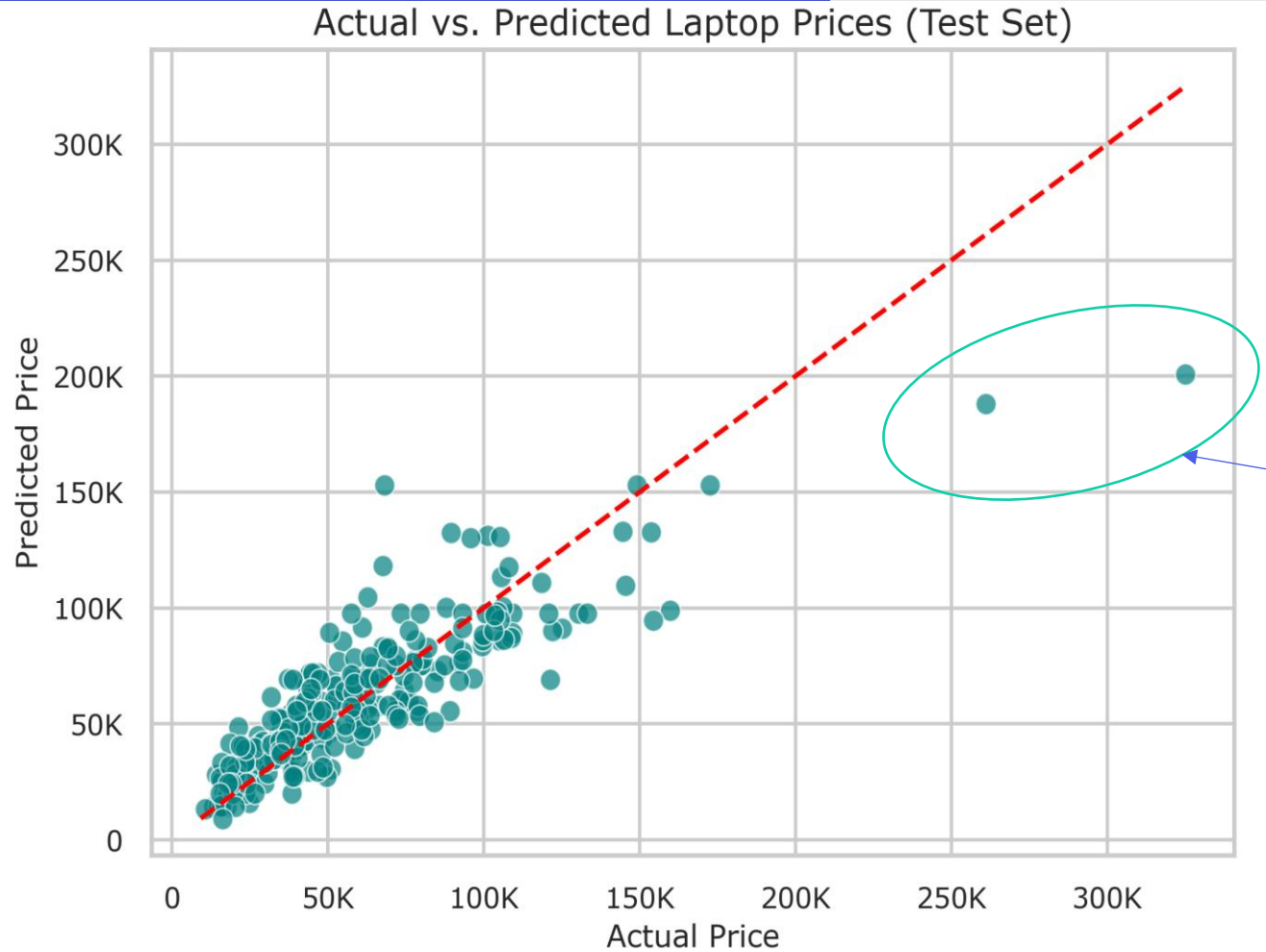
* Looking at Coefficient of the model Ram affecting price by 14.395,82 point almost double SSD at 8.000,63 point & triple Pixels at 5.349,26 point

THE MODELS PERFORMANCE



- The tight cluster of green dot following the red line, depict the model could works beautifully on predicting cheaper laptop.
- On average the model is off by about 13,638.00

THE MODELS PERFORMANCE



- Although, Looking at outlier on the right-side pricier laptop are harder to predict. This is due to the lack of luxurious laptop data on the dataset.

BUSINESS VALUE

How can this model be used?

- Automatically spotting underpriced laptops for resale.
- helping manufacturers price new models competitively.



THANK YOU

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